

CHAMBERLAIN, SD, USA

WMO: 726530

Lat: 43.767N

Lon: 99.318W

Elev: 1703

StdP: 13.81

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 94943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-3.0	-14.8	2.6	-7.1	-10.2	3.4	-2.5	31.2	21.9	27.8	21.4	8.0	280	0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	96.6	72.2	92.4	71.6	89.1	70.8	77.5	87.4	75.6	85.8	73.8	84.2	13.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	140.3	82.3	72.8	129.8	80.7	70.6	120.1	79.4	42.4	87.2	40.4	85.8	38.5	83.7	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.9	25.0	22.2	-17.3	103.2	5.3	5.3	-21.1	107.1	-24.2	110.2	-27.2	113.2	-31.1	117.0		
			-17.7	79.8	5.2	2.5	-21.4	81.6	-24.5	83.0	-27.4	84.4	-31.2	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	21.1	22.8	36.7	47.7	58.4	69.4	76.3	73.1	64.9	50.2	36.6	23.5
	DBStd	21.85	13.27	12.94	13.02	10.25	8.81	6.69	6.54	6.15	8.34	9.74	10.98	11.70
	HDD50	3642	896	762	442	158	25	0	0	0	3	119	415	822
	HDD65	6940	1361	1181	879	524	241	34	4	9	102	467	851	1287
	CDD50	3104	0	1	29	88	285	582	814	716	450	125	14	0
	CDD65	927	0	0	1	4	36	165	353	260	99	9	0	0
	CDH74	10106	0	0	26	100	462	1676	3982	2574	1131	151	4	0
	CDH80	4345	0	0	4	24	149	619	1959	1117	433	40	0	0
Wind	WSAvg	10.4	10.4	10.3	11.5	12.1	11.3	10.2	9.3	9.1	10.2	10.5	9.9	10.0
Precipitation	PrecAvg	22.1	0.4	0.6	1.0	2.5	3.4	4.4	2.4	2.5	2.1	1.8	0.7	0.4
	PrecMax	29.2	1.7	1.2	2.4	6.3	6.4	8.2	5.7	6.3	6.8	6.5	1.8	1.5
	PrecMin	14.3	0.0	0.0	0.0	0.5	0.5	1.7	0.2	0.1	0.4	0.0	0.0	0.0
	PrecStd	4.3	0.4	0.3	0.7	1.4	1.8	1.7	1.6	1.4	1.5	1.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	62.3	78.3	83.5	90.4	94.6	105.5	99.8	96.0	85.4	72.3	56.3
		MCWB	43.3	48.1	56.4	58.9	67.6	72.6	71.8	72.0	68.8	61.7	53.3	43.4
	2%	DB	47.6	52.2	69.2	76.0	84.1	90.5	98.7	93.7	90.1	78.2	63.7	48.6
		MCWB	39.0	42.2	52.3	57.3	65.5	71.6	73.0	71.5	68.8	60.0	49.4	39.6
	5%	DB	42.8	45.4	62.7	70.3	79.1	86.9	93.8	90.0	85.2	72.2	57.7	43.2
		MCWB	36.6	38.8	49.9	54.2	63.2	70.2	72.8	71.0	66.5	57.2	46.2	36.7
	10%	DB	38.7	39.8	56.2	65.3	74.2	83.3	90.0	86.2	81.0	66.5	52.5	38.5
		MCWB	34.3	34.8	46.0	51.7	61.1	68.2	72.1	69.9	65.1	54.0	43.5	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	49.5	59.3	62.4	71.7	80.4	81.3	78.3	75.3	65.7	55.0	44.7
		MCDB	54.8	61.5	67.3	76.3	84.8	88.5	89.8	87.6	83.2	79.8	68.0	53.6
	2%	WB	39.5	43.0	54.1	59.1	68.1	75.9	78.2	76.2	72.5	61.8	50.9	40.5
		MCDB	45.7	51.3	65.9	73.1	80.0	84.7	87.7	85.8	82.3	73.6	62.4	48.1
	5%	WB	37.0	38.7	50.4	55.8	65.5	72.7	76.4	74.4	69.9	58.4	47.1	37.3
		MCDB	42.1	45.2	61.9	67.2	76.0	82.3	87.2	84.2	80.6	70.6	57.2	42.4
	10%	WB	34.6	35.4	46.3	53.0	62.5	70.4	74.5	72.3	67.6	55.0	43.8	34.4
		MCDB	38.6	39.7	56.1	63.3	71.8	80.2	85.8	82.4	77.6	64.8	51.8	38.3
Mean Daily Temperature Range	5% DB	MDBR	18.4	19.5	21.3	22.0	21.6	21.7	23.7	22.7	24.0	22.4	21.0	17.1
		MCDBR	23.9	26.5	32.9	32.4	28.7	26.5	28.7	28.1	30.4	31.5	29.7	23.9
		MCWBR	17.3	17.8	18.5	17.0	14.1	11.8	10.1	10.8	12.7	16.2	17.3	16.6
	5% WB	MCDBR	22.9	25.6	30.6	28.6	24.6	22.3	23.5	22.5	24.6	28.3	29.0	22.9
		MCWBR	17.0	17.3	17.8	16.3	13.5	12.3	10.9	11.4	12.3	15.7	17.0	16.4
Clear-Sky Solar Irradiance	taub	0.263	0.279	0.304	0.344	0.366	0.377	0.392	0.392	0.350	0.316	0.277	0.263	
	taud	2.498	2.474	2.477	2.360	2.382	2.391	2.374	2.356	2.444	2.509	2.533	2.494	
	Ebn at Noon	282	294	299	292	287	283	277	273	278	276	273	270	
	Edh at Noon	22	26	30	37	37	37	37	37	31	25	21	20	
All-Sky Solar Radiation	RadAvg	601	913	1235	1560	1850	2090	2164	1815	1452	974	656	488	
	RadStd	49	67	106	115	169	147	77	143	97	117	64	37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page